DAY 08 · BEGINNER · FREE API

Your First Agent

Same tools, but now the model decides what to do next, and how many times round the loop to go.

create_agent | the trace | stream | call limits | Notebook

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The 10-Day Course Map

WEEK 1 TALKING TO A MODEL

01 Your First AI Program
Ask a model, print the answer

02 Roles and Messages
The system message sets the job

03 Prompt Templates
One template, many inputs

YOU WILL HAVE BUILT
a script that answers anything you type into it

WEEK 2 BUILDING SOMETHING

04 Chains: The Pipe
`prompt | model | parser`

05 Structured Output
An object, not a paragraph

06 Memory
A checkpointer and a thread id

YOU WILL HAVE BUILT
a chatbot that remembers what you said a minute ago

WEEK 3 GIVING IT ABILITIES

07 Tools
Your Python, called by the model

YOU ARE HERE

08 Your First Agent
It decides when to call them

09 Chunks and Vectors
Find a paragraph by meaning

10 RAG with RAGWire
Citations, and a refusal

One notebook and one deck a day, **60 to 90 minutes** each. Every model used is a **free** one, and Docker only turns up on Day 9.

A MODEL, A TOOL LIST AND A SYSTEM PROMPT

Building an Agent: Three Arguments

02

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`model`

the one that decides

`tools=[to_rupees, multiply]`

what it may use

`system_prompt`

"Always use a tool for any calculation."

"I am buying 12 items at 250 dollars each. How many rupees is that?"

Two steps happened, and **you named neither of them.**

Multiply first, then convert. **Nothing in your code said so.**

Reading the Agent Trace

HUMAN	"12 items at 250 dollars each. How many rupees?"	
AI	`multiply(a=12, b=250)`	decides
TOOL	`3000.0`	computed by Python
AI	`to_rupees(amount=3000, currency='USD')`	decides again
TOOL	`265200.0`	
AI	"The total is $3,000, which converts to ₹265,200."	

Yesterday's loop exactly. The only new thing is **who decides when to stop**.

Chain vs Agent: The Cost

A CHAIN

one model call

The steps you planned,
in the order you planned.

Use when you know the steps

AN AGENT

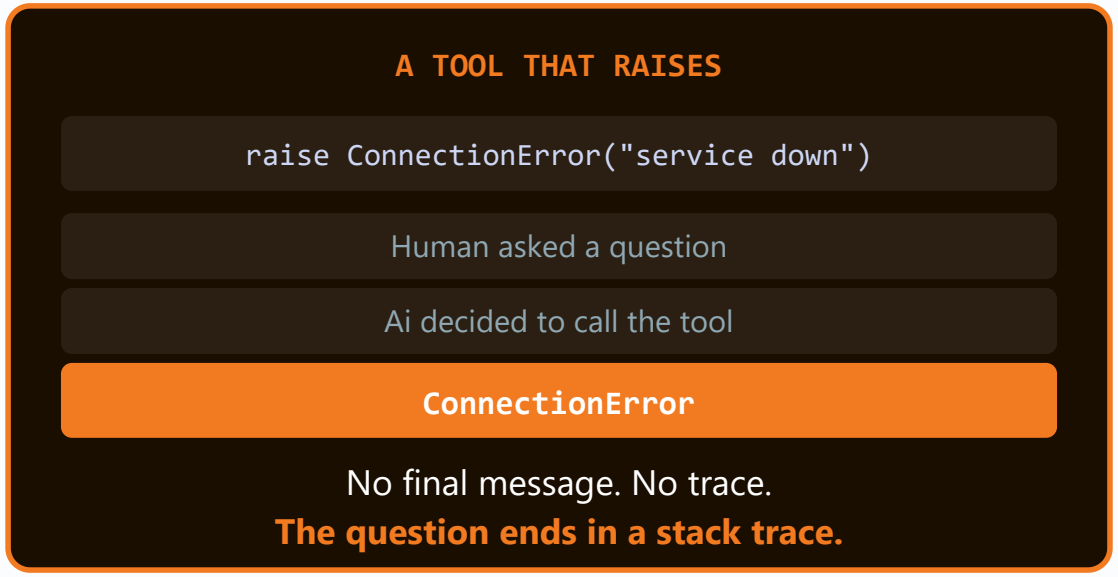
1,713 tokens for the two-step question

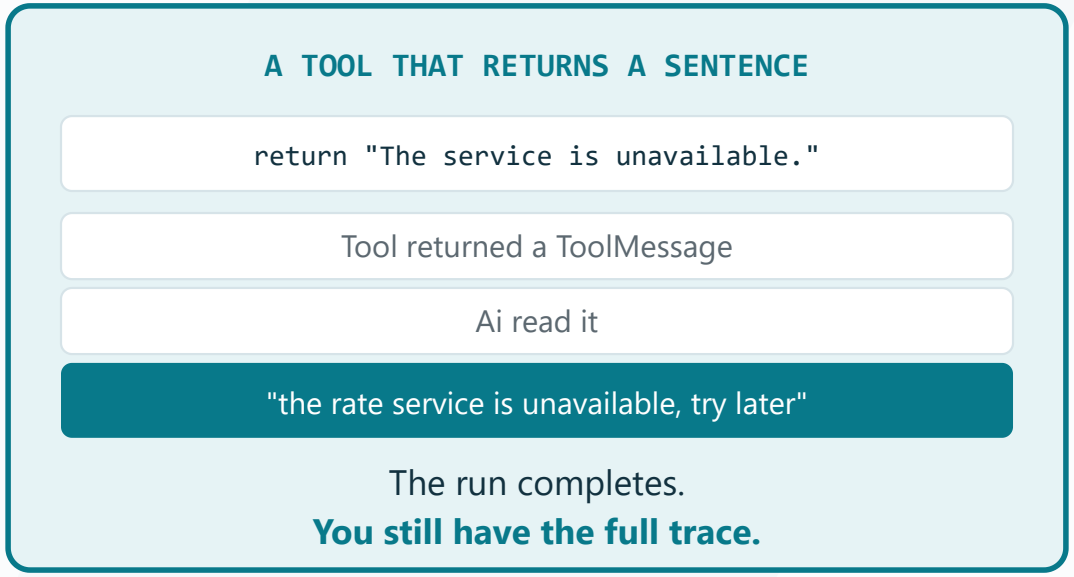
Even "say hello in one word"
costs two messages.

Use when the next step depends

An agent buys you **decisions**, and decisions cost round trips.

Handling Tool Failures





An agent is only as reliable as its **least defensive tool**. The fix goes in the tool.

Tomorrow: Documents and Vectors

Your agent can calculate, convert, remember and explain itself.
Days 1 to 8, in one object

It still cannot answer the second question from Day 7.
"How many kg of checked baggage does a Zephyr economy fare include?"

That answer lives in a
document on your disk

Tomorrow: take documents apart
and search them by meaning

No keyword search. Chunks, vectors, and one line of arithmetic.